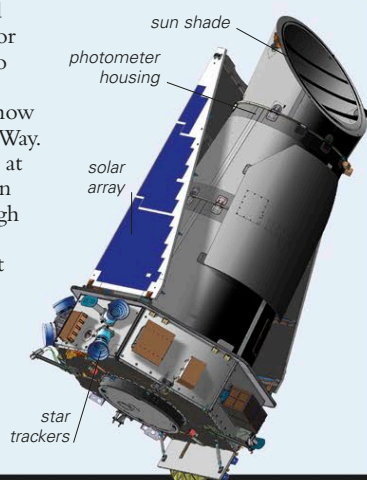


EXPLORING SPACE

THE KEPLER MISSION

NASA's Kepler spacecraft was designed to carry out the first extensive search for transiting exoplanets, leading not just to the discovery of individual planets, but also to the first statistical evidence for how many planets might exist in the Milky Way. Kepler's primary mission was to "stare" at a single patch of sky in the constellation Cygnus, directing light collected through its 3.1-ft (0.95-m) telescope onto an advanced photometer (light meter) that continuously recorded the brightness of 150,000 main-sequence stars. When the pointing system that kept Kepler precisely aligned to its target area began to fail in 2013, the mission was redesigned to switch to target a new field of view every few months.



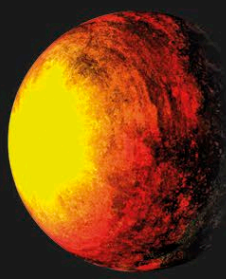
PROXIMA
CENTAURI b



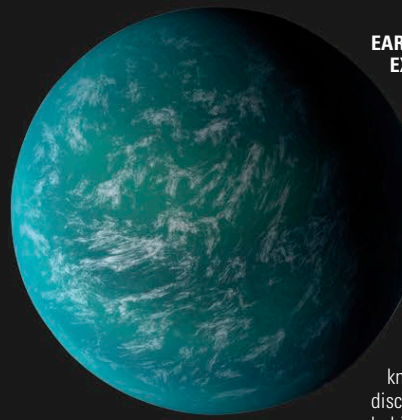
KEPLER-37b



EARTH



KEPLER-10b



KEPLER-22b

EARTH AMONG THE EXOPLANETS

Earth is shown here to scale with some significant rocky exoplanets. Proxima Centauri b is the closest exoplanet to Earth, while Kepler-37b is one of the smallest worlds so far discovered. Kepler-10b is a searing-hot world that was the first to be discovered by Kepler, while Kepler-22b was the first known transiting exoplanet discovered within its star's habitable zone, potentially supporting a watery surface.

PROXIMA CENTAURI b

In 2016, astronomers announced the discovery of a roughly Earth-sized planet orbiting Proxima Centauri, our nearest stellar neighbor. Although technically in the habitable zone of its red dwarf star, Proxima b orbits at a distance of just 4.7 million miles (7.5 million km), putting it in the firing line of frequent violent flares and lessening the chances of this being a truly Earth-like world on our doorstep.



TRAPPIST-1 SYSTEM

In 2015, three roughly Earth-sized planets were discovered orbiting the red-dwarf star TRAPPIST-1, 39.6 light-years away. A further four have joined them since, making the system a prime target in the hunt for truly Earth-like worlds. Three of the known planets orbit within the habitable zone, with planets d and e thought most likely to support an Earth-like climate.



ROCKY EXOPLANETS

From around 2005 onward, advances in the transit method (particularly using dedicated satellites such as Kepler) have seen the number of smaller planets catch up with and eventually overtake the known gas giants. The masses of these smaller, rocky worlds can be estimated from their diameter (the strength of the transit) and assumptions about their density and composition—they rarely have enough mass to affect the radial velocity of their parent star. Known rocky exoplanets range from Moon-sized worlds to super-Earths with several times our planet's mass, and even a couple of "mega-Earths" whose masses must approach those of Neptune. Missions such as NASA's Transiting Exoplanet Survey Satellite (TESS) and ESA's CHEOPS continue to expand our understanding of these intriguing worlds.

LOOKING FOR EARTHS

If there is life elsewhere in the universe, it seems reasonable to expect to find it on a world similar to Earth: a rocky planet orbiting a main-sequence star. Evidence from the Kepler mission and other search programs suggests that about 20 percent of Sun-like stars have at least one giant planet. Current techniques can be used to estimate the orbital parameters of gas giants around these stars and thereby identify those with stable, circular orbits at a fair distance from their respective stars. Within a significant proportion of such systems, there is likely to be an inner zone where rocky terrestrial planets may have formed, some of them within the habitable zones associated with those stars. Although to date no exoplanet perfectly resembling Earth has been found, the signs are encouraging that one or more may be found in the reasonably near future. Once detected, analysis of light reflecting from the planets' atmospheres for telltale signs of life, such as oxygen and methane, should be possible—a capacity for this is already present in some existing telescopes involved in the exoplanet search.

HABITABLE ZONE

For life to develop on a planet, the planet must lie within the host star's "habitable zone," where liquid water can permanently exist on the surface. This zone's extent depends on the star's mass and luminosity.

